Lecture 16: Summarizing Data and Managing Missing Data

Heather Mattie, PhD

October 22, 2026

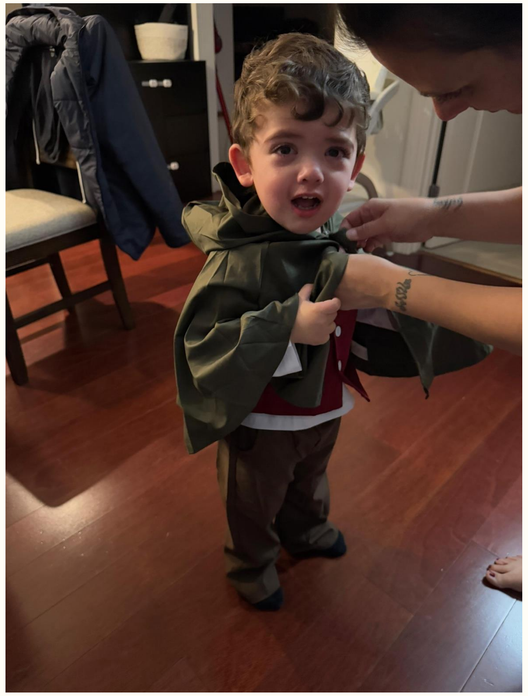
Recipe of the Day!

Turkish Çiğ Köfte



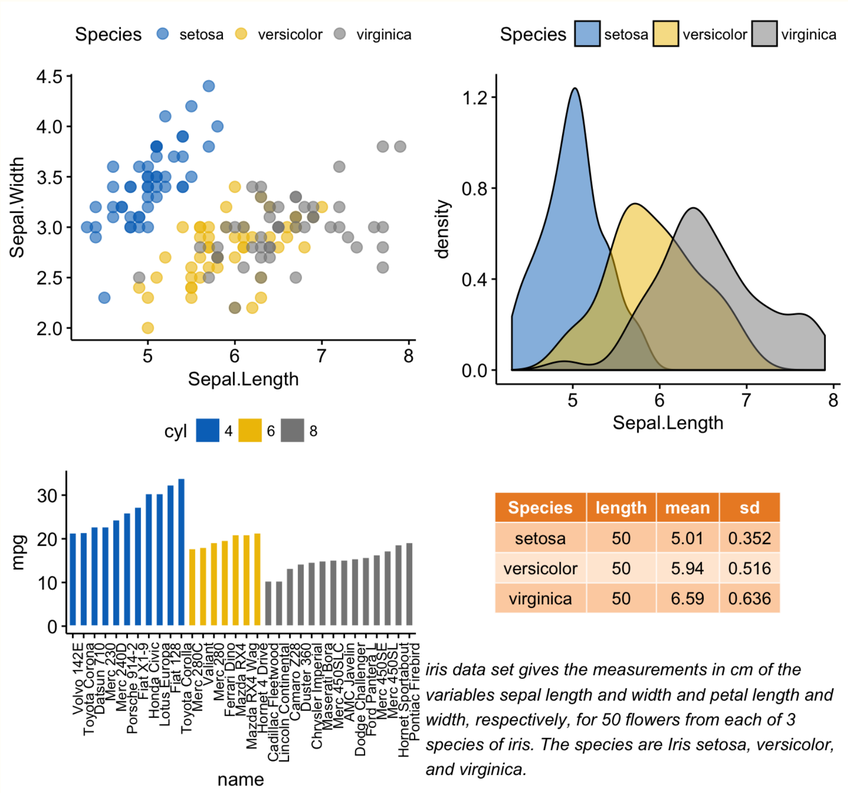






Announcements

- Homework #3 due tomorrow, October 23 by 11:59pm
- Heather's office hours this week
  - Today 12-1pm
- No lab this week
- Midterm will be released on **Friday, October 23** and due **Friday, November 6**
  - No late days may be used



Species	length	mean	sd
setosa	50	5.01	0.352
versicolor	50	5.94	0.516
virginica	50	6.59	0.636

iris data set gives the measurements in cm of the variables sepal length and width and petal length and width, respectively, for 50 flowers from each of 3 species of iris. The species are Iris setosa, versicolor, and virginica.

Combining plots and tables

Midterm details

- Will be released tomorrow, October 23rd
- Due Friday, November 6th
- You may ask me or the TFs questions via email
  - All questions and answers will be added to a Google doc so everyone has access to the same information (just Q&A, no student names)
  - The link to the Google doc will be in the Midterm and on the Midterm assignment page in Canvas
  - It is your responsibility to check if there have been any updates to the doc
- Read each question carefully and make sure you answer all parts of a question

Midterm details

RULES FOR THIS EXAM

- **Allowed:** lecture slides and notes, homework solutions, lab solutions, and online documentation (e.g., function help pages).
- **Not allowed:** generative AI in **any** form — this includes using AI to debug code or to explain how a function works.
- **Individual work only.** Communicating with other students about the exam is **not** permitted.
- **Questions?** Email Dr. Mattie and the teaching fellows. All questions and answers will be posted to this Google Doc so everyone has the same information. **It is your responsibility to check this document while the exam is open.**
- **No late days** are available for this exam unless you have accommodations through the Office of Student Affairs. If an emergency prevents you from finishing on time, email the instructor as soon as possible.
- This exam may not be shared with any individual, group, or company, or posted anywhere online. Doing so violates the School's Academic Integrity Policy as stated in the Student Handbook, and students who distribute or use course material in this way are subject to disciplinary action.

Coding question of the day

The **`ToothGrowth`** dataset records odontoblast length (**`len`**) in 60 guinea pigs given vitamin C at three doses (**`dose`**: 0.5, 1, or 2 mg/day) by one of two delivery methods (**`supp`**: orange juice OJ or ascorbic acid VC). Make a boxplot of tooth length by dose, with boxes filled by delivery method. Add informative axis labels and a title.

Extension: Try **`facet_wrap(~ supp)`** instead of **`fill`**. Which version makes the comparison easier to see?

Be sure to run this code first

```
library(tidyverse)
```

Summarizing data

Reporting missing data

Heather Mattie, PhD

Missing data

Missing data is a common issue in biomedical datasets

- Can bias results and reduce statistical power
- Some models and functions will not run if there is missing data
- Clear communication of missingness improves reproducibility and transparency



`gapminder` example

```
library(tidyverse)
library(dslabs)
data(gapminder)

df <- gapminder %>%
  filter(year == 1965)

head(df)
```

```
##               country year infant_mortality life_expectancy fertility
## 1             Albania 1965            94.10           66.59      5.59
## 2             Algeria 1965           149.10           50.09      7.66
## 3              Angola 1965               NA           38.74      7.43
## 4 Antigua and Barbuda 1965               NA           65.23      4.19
## 5           Argentina 1965            59.11           65.95      3.06
## 6             Armenia 1965               NA           69.26      3.98
##   population          gdp continent          region
## 1    1896125           NA    Europe Southern Europe
## 2   12626953  14486635612    Africa Northern Africa
## 3    5765025           NA    Africa   Middle Africa
## 4      59653           NA  Americas       Caribbean
## 5   22283389 130560392722  Americas   South America
## 6    2204650           NA      Asia    Western Asia
```

is.na

The `is.na` function returns TRUE if a value is NA (missing) and FALSE if it is not NA

- Using `is.na` with the `sum` function shows how many values are NAs in an object

```
sum(is.na(df))
```

```
## [1] 149
```

```
sum(is.na(df$country))
```

```
## [1] 0
```

```
sum(is.na(df$year))
```

```
## [1] 0
```

```
sum(is.na(df$infant_mortality))
```

```
## [1] 66
```

```
sum(is.na(df$life_expectancy))
```

```
## [1] 0
```

```
sum(is.na(df$fertility))
```

```
## [1] 0
```

```
sum(is.na(df$population))
```

```
## [1] 0
```

```
sum(is.na(df$gdp))
```

```
## [1] 83
```

```
sum(is.na(df$continent))
```

```
## [1] 0
```

```
sum(is.na(df$region))
```

```
## [1] 0
```

colSums

The `colSums` function with `is.na` returns the number of NAs in each column

```
colSums(is.na(df))
```

```
##          country             year infant_mortality  life_expectancy
##                0                0               66                0
##        fertility       population              gdp        continent
##                0                0               83                0
##           region
##                0
```

rowSums

The `rowSums` function with `is.na` returns the number of NAs in each row

```
rowSums(is.na(df))
```

```
##   [1] 1 0 2 2 0 2 2 0 0 2 1 1 0 0 2 0 1 0 2 0 2 0 0 2 1 0 0 2 0 0 2 0 1 0 0 0 2
##  [38] 1 0 0 0 2 1 2 2 0 2 0 0 0 0 2 2 2 2 0 0 0 1 1 1 1 2 0 0 2 2 0 1 2 0 1 0 1
##  [75] 0 0 0 0 0 1 1 1 0 1 0 1 2 0 1 0 1 2 2 1 1 0 0 1 2 0 2 2 0 0 0 1 1 1 0 1 0
## [112] 2 2 2 2 0 1 2 0 0 1 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 2 2 2 0 2 1 2 2 0 2 0 0
## [149] 0 2 2 1 1 0 0 0 2 1 0 1 0 2 1 0 2 0 1 0 0 0 2 1 2 1 0 0 0 2 1 0 2 1 1 0 0
```

`skimr` and `skim`

The `skimr` package provides compact and flexible ways to generate summary statistics

- The `skim` function outputs a comprehensive data summary

```
library(skimr)
skim(df)
```

Data summary

Name	df
Number of rows	185
Number of columns	9
Column type frequency:	
factor	3
numeric	6
Group variables	None

`skimr` and `skim`

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
country	0	1	FALSE	185	Alb: 1, Alg: 1, Ang: 1, Ant: 1
continent	0	1	FALSE	5	Afr: 51, Asi: 47, Eur: 39, Ame: 36
region	0	1	FALSE	22	Wes: 18, Eas: 16, Wes: 16, Car: 13

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
year	0	1.00	1.965000e+03	0.000000e+00	1965.00	1.96500e+03	1.965000e+03	1.965000e+03	1.965000e+03	
infant_mortality	66	0.64	9.711000e+01	5.817000e+01	13.70	5.15500e+01	9.620000e+01	1.372000e+02	2.636000e+02	
life_expectancy	0	1.00	5.726000e+01	1.109000e+01	31.26	4.84400e+01	5.834000e+01	6.705000e+01	7.400000e+01	
fertility	0	1.00	5.380000e+00	1.830000e+00	1.81	3.60000e+00	6.070000e+00	6.800000e+00	8.200000e+00	
population	0	1.00	1.759520e+07	6.650895e+07	37963.00	7.53285e+05	3.559687e+06	9.504702e+06	7.065909e+08	
gdp	83	0.55	7.915212e+10	3.387851e+11	110094728.00	1.41424e+09	4.138937e+09	2.746349e+10	3.162743e+12	

`naniar`

The `naniar` package is a `tidyverse`-friendly package designed to facilitate the exploration, visualization, and manipulation of missing data.

- It provides a suite of tools for understanding the patterns and mechanisms of missingness within a dataset



gg_miss_var

The **gg_miss_var** function is used to visualize the missing data in a dataset by variable.

- Generates a **ggplot2** plot that displays the number of missing values for each variable (column) in a data frame

```
library(naniar)

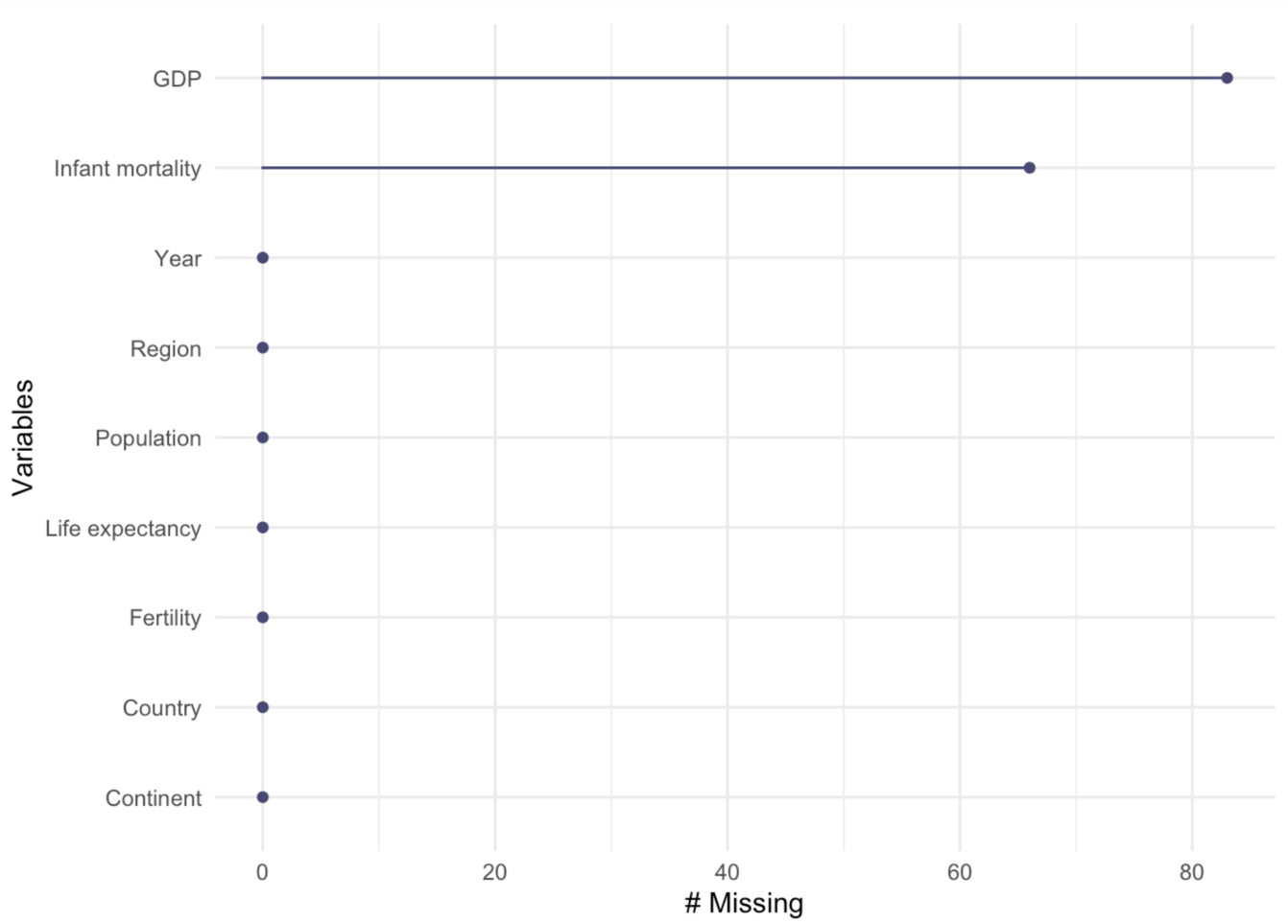
df <- df %>%
  rename("Infant mortality" = infant_mortality,
         "Life expectancy" = life_expectancy,
         "Population" = population,
         "Fertility" = fertility,
         "Continent" = continent,
         "Country" = country,
         "Region" = region,
         "Year" = year,
         "GDP" = gdp
         )

gg_miss_var(df)
```

gg_miss_var

The `gg_miss_var` function is used to visualize the missing data in a dataset by variable.

- Generates a `ggplot2` plot that displays the number of missing values for each variable (column) in a data frame

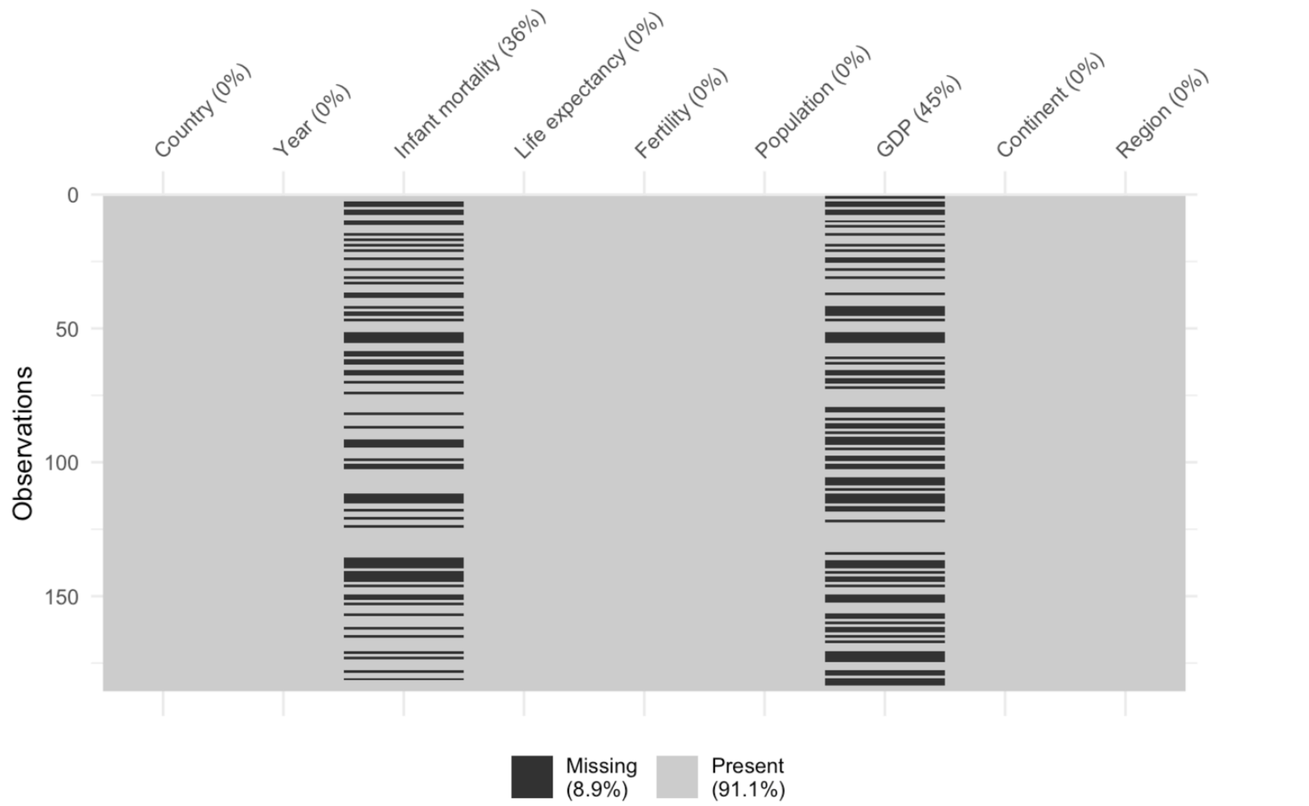


vis_miss

The **vis_miss** function allows for the visualization of missing data patterns within a data frame.

- Generates a **ggplot** object, offering a quick and visual overview of where missing values and present values are located within the dataset.

```
vis_miss(df)
```



Summary

- Missing data is a common issue and should be reported
- The `colSums` and `rowSums` can be used to show the number of missing values in columns and rows
- The `skim` function can be used to generate a comprehensive missing data summary
- Missing data can be visualized by using the `gg_miss_var` and `vis_miss` functions

Managing missing data

Heather Mattie, PhD

HARVARD T.H. CHAN
SCHOOL OF PUBLIC HEALTH

Managing Missing Data Roadmap

1. Types of missingness
2. Simple imputation methods
3. Advanced imputation methods
4. Survey skip patterns

Managing missing data

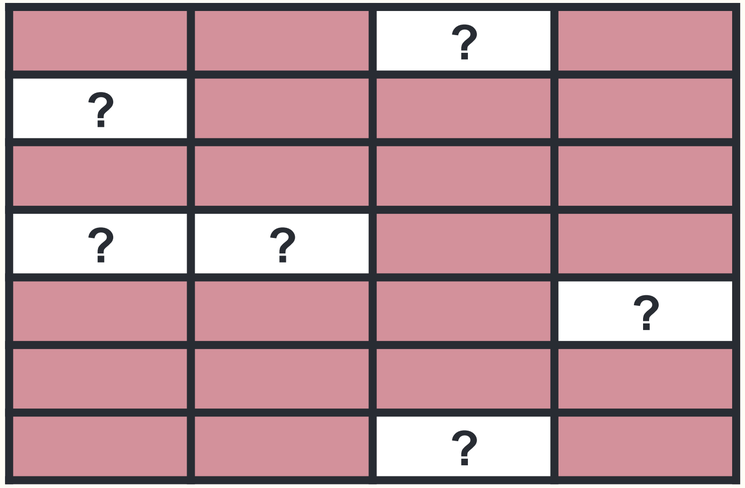
Types of missingness

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Missingness mechanisms

There are four types of missingness

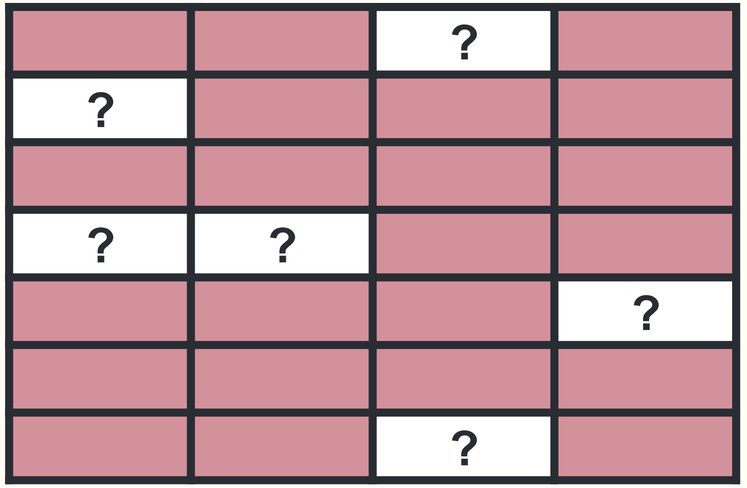
1. Missing completely at random (MCAR)
2. Missing at random (MAR)
3. Not missing at random (NMAR)
4. Missingness by design



Understanding patterns of missingness is a crucial first step in handling missing values

MCAR

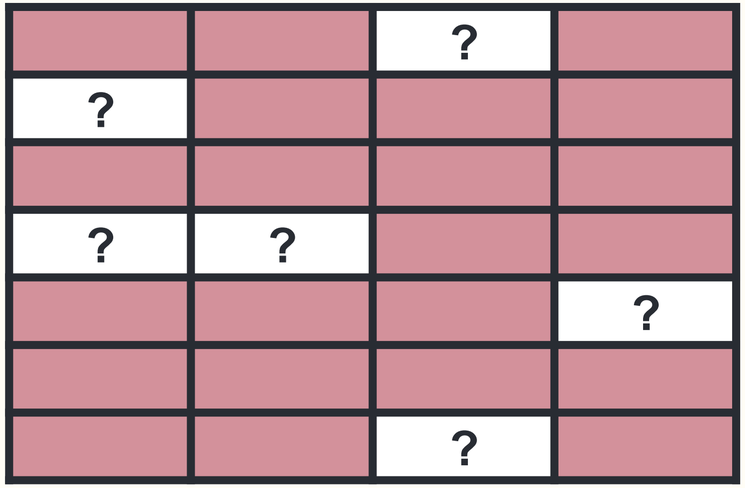
Missing completely at random (MCAR) means that there is no relationship between the probability of data being missing and any of the other variables in the data



- The probability of being missing is the same for all cases
- Rare in public health research

MCAR

- **Example**: a patient's blood pressure reading missing due to device malfunction, regardless of other patient information
- **Solution**: simple imputation methods (mean, median, mode, LOCF, BOCF)



MAR

Missing at random (MAR) means that the data is missing in a systematic, predictable way based on the other information (variables) in the dataset

- Missingness is not related to the value that is missing itself
- Common in public health research



MAR

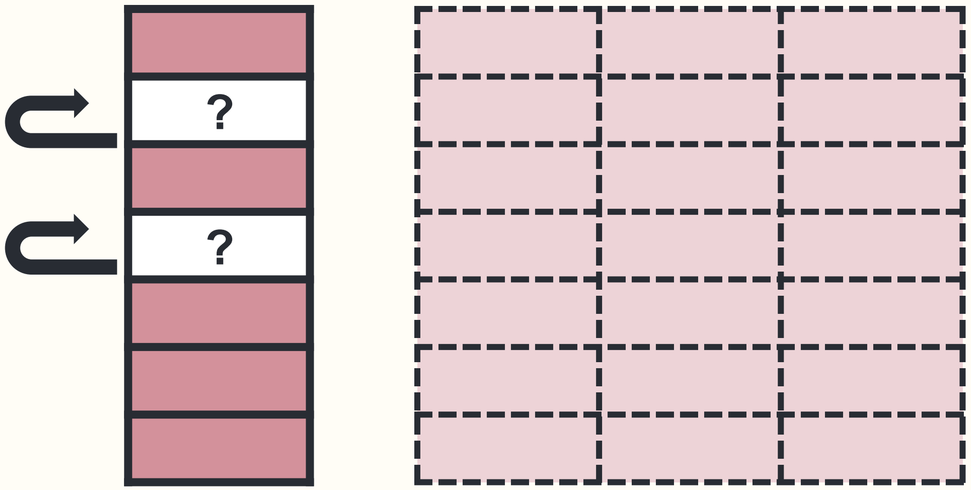
- Example: the presence of a fever missing because all patients with chills and aches were assumed to have a fever and their temperature was not taken
- Solution: multiple imputation



NMAR

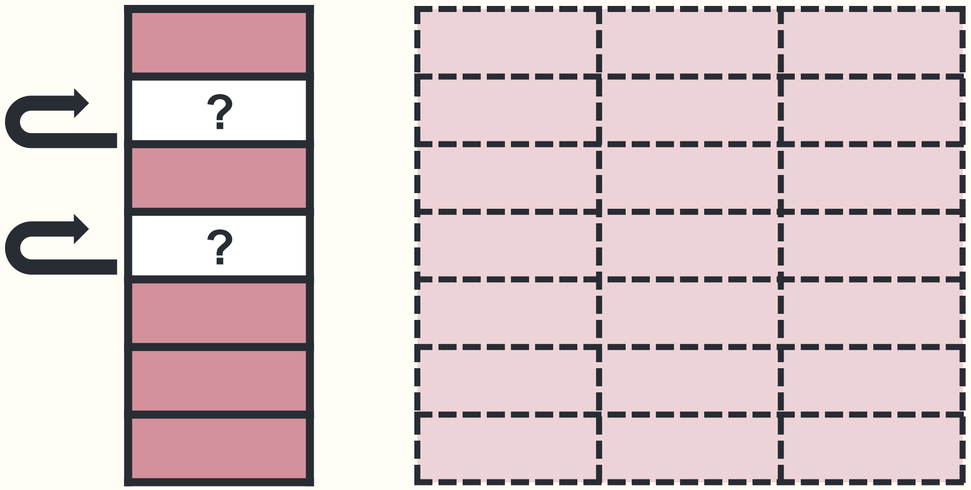
Not missing at random (NMAR) means that the data is NOT missing in a systematic, predictable way based on the other information (variables) in the dataset, but also isn't missing randomly

- Missingness is attributed to the missing value itself or the variables not collected
- There is a pattern to the missingness, but it isn't due to any observed data



NMAR

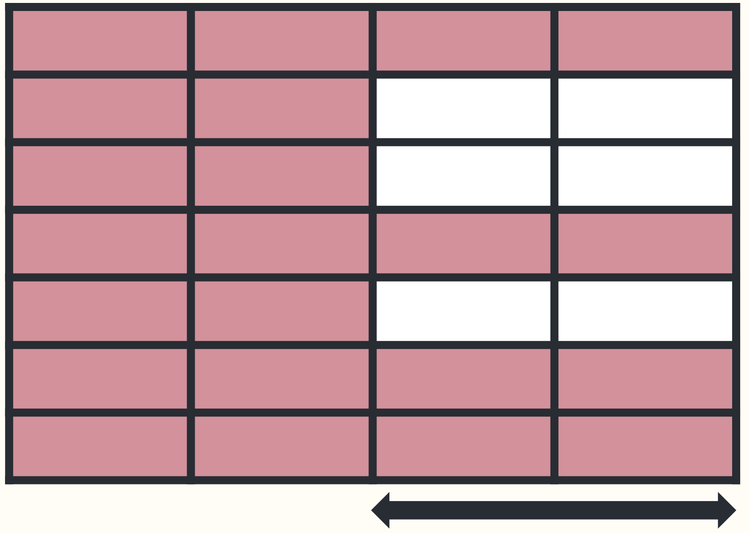
- **Example**: the age of very elderly patients missing because they don't know it or refuse to say how old they are
- **Solution**: Try to collect more data or information about why data is missing; imputation is usually not recommended



Missingness by design

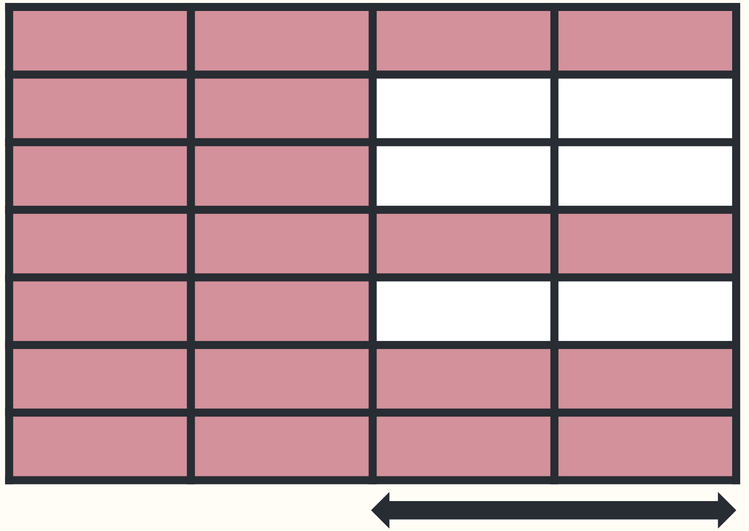
Missingness by design refers to intentionally leaving out data according to a predefined plan

- Common in surveys
- Often used to reduce participant burden or data collection costs
- Missing values can typically be imputed in a straightforward way



Missingness by design

- **Example**: respondents of a survey skipping questions about smoking habits if they are non-smokers
- **Solution**: skip pattern imputation



Summary

- There are four types of missingness: MCAR, MAR, NMAR, and missingness by design
- Ignoring the missingness mechanism can lead to biased or misleading results
- Each type of missingness should be handled in a different way

Managing missing data

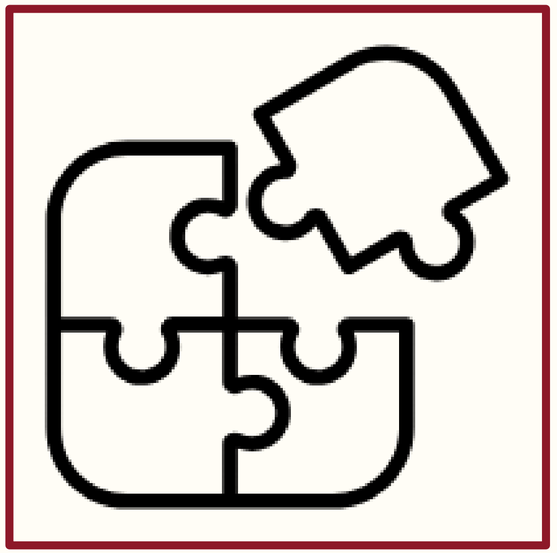
Simple imputation methods

Heather Mattie, PhD

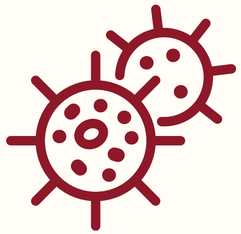
Simple imputation methods

When it can be assumed the type of missingness is MCAR, simple imputation methods can be used

- Methods we will cover include mean, median, mode, LOCF, and BOCF
- MCAR is a strong assumption and simple imputation methods should be used with caution



Simulated Ebola outbreak dataset



Simulated data mimicking the West African Ebola outbreak of 2014-2015

- **`fever`**: indicator of fever
- **`temp`**: temperature in degrees Celsius
- **`outcome`**: indicator of recovery or death

```
ebola_subset <- ebola %>%
  select(temp, fever, outcome)

head(ebola_subset)
```

```
##   temp fever outcome
## 1 36.8    no    <NA>
## 2 36.9  <NA> Recover
## 3 36.9  <NA> Recover
## 4 36.8    no    <NA>
## 5 36.9    no Recover
## 6 37.6    no Recover
```

```
colSums(is.na(ebola_subset))
```

```
##    temp   fever outcome
##     149     249    1323
```

Mean imputation

Mean imputation replaces missing values in a column with the mean of the non-missing values in the same column

- For numeric values only
- Can use the **`mean`** function

```
ebola <- ebola %>%
  mutate(temp_replace_na_with_mean =
           replace_na(temp, mean(temp, na.rm = TRUE)))

mean(ebola$temp, na.rm = TRUE)
```

```
## [1] 38.55829
```

```
mean(ebola$temp_replace_na_with_mean)
```

```
## [1] 38.55829
```

Median imputation

Median imputation replaces missing values in a column with the median of the non-missing values in the same column

- For numeric or categorical values
- Can use the **`median`** function

```
ebola <- ebola %>%
  mutate(temp_replace_na_with_median =
           replace_na(temp, median(temp, na.rm = TRUE)))

median(ebola$temp, na.rm = TRUE)
```

```
## [1] 38.8
```

```
median(ebola$temp_replace_na_with_median)
```

```
## [1] 38.8
```

Mode imputation

Mode imputation replaces missing values in a column with the mode of the non-missing values in the same column

- Primarily used for categorical variables
- Can use the **`impute_mode`** function from the **`missMethods`** package

```
library(missMethods)

table(ebola$fever, useNA = "ifany")
```

```
##
##   no  yes <NA>
## 1090 4549  249
```

```
ebola$fever <- impute_mode(as.data.frame(ebola$fever))

table(ebola$fever, useNA = "ifany")
```

```
## ebola$fever
##   no  yes
## 1090 4798
```

LOCF and BOCF

Last observation carried forward (LOCF) and baseline observation carried forward (BOCF) are imputation methods for handling missing data in longitudinal studies

- LOCF replaces a missing value with the last recorded (observed) value for that variable for a particular person
- BOCF replaces all missing values of a participant's variable with the baseline (first) measurement

LOCF and BOCF

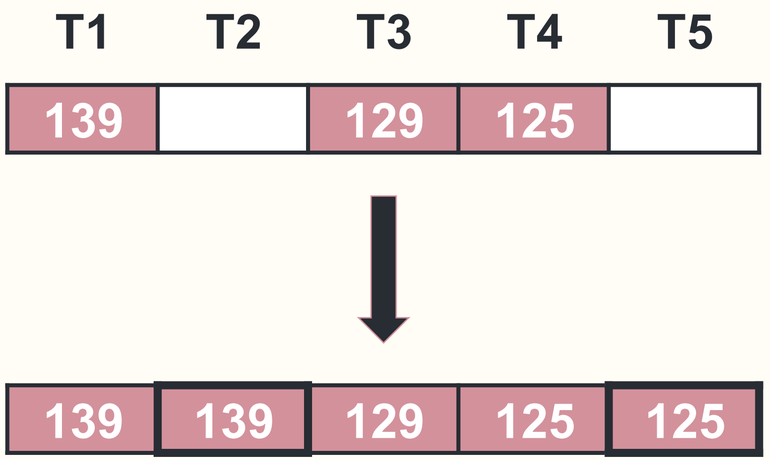
Last observation carried forward (LOCF) and baseline observation carried forward (BOCF) are imputation methods for handling missing data in longitudinal studies



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LOCF and BOCF

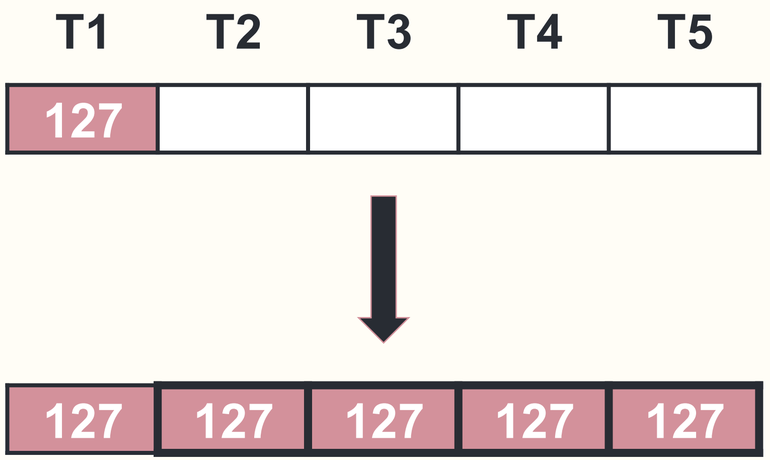
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LOCF and BOCF

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- BOCF replaces all missing values of a participant's variable with the baseline (first) measurement

LOCF and BOCF

```
data
```

```
##    patID visit sbp
## 1      1     1 139
## 2      1     2  NA
## 3      1     3 129
## 4      1     4 125
## 5      1     5  NA
## 6      2     1 127
## 7      2     2  NA
## 8      2     3  NA
## 9      2     4  NA
## 10     2     5  NA
```



```
data <- data %>% fill(sbp)
print(data)
```

```
##    patID visit sbp
## 1      1     1 139
## 2      1     2 139
## 3      1     3 129
## 4      1     4 125
## 5      1     5 125
## 6      2     1 127
## 7      2     2 127
## 8      2     3 127
## 9      2     4 127
## 10     2     5 127
```

Summary

- Simple imputation methods can be used when the data missingness mechanism is MCAR
- Mean imputation and median imputation are common methods for numerical data
- Mode imputation is common for categorical data
- LOCF and BOCF are used for longitudinal data